

HEMANT BAGHEL

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Agra, Uttar Pradesh, India

PROFESSIONAL SUMMARY

Computer Vision / AI-ML Engineer Intern with hands-on experience developing, testing, and deploying real-time AI systems for defence technology applications. Experienced in aircraft detection and classification, object detection, image processing, deep learning inference, dataset preparation, camera-based systems, and edge AI deployment. Currently working with NVIDIA Jetson, Linux/Ubuntu, workstations, RTSP/IP-stream cameras, thermal cameras, PLC integration, TCP/IP and Ethernet communication in fully offline environments. Experience with YOLO-based detection, ConvNeXt-based classification, confidence/OOD validation, real-time monitoring, logging, recording, and system integration. Previously completed AI internship at ADRDE (DRDO) and a research internship at IIT Jammu focused on computer vision, adversarial robustness, and AI security.

TECHNICAL SKILLS

Programming: Python, C++, SQL

Computer Vision: OpenCV, YOLOv5/v8, Object Detection, Image Processing, Object Tracking, ArUco Markers, Camera Processing, Thermal Imaging

AI / Machine Learning: Machine Learning, Deep Learning, Model Training, Image Classification, Feature Extraction, Confidence Analysis, OOD Detection, Model Evaluation

Deep Learning: PyTorch, YOLO, ConvNeXt, Neural Network Inference

Libraries / Frameworks: NumPy, Pandas, Scikit-learn, Flask

Edge AI / Deployment: NVIDIA Jetson, Linux, Ubuntu, Real-Time Inference, Edge AI, Workstations, Offline AI Deployment

Camera / Industrial Systems: RTSP, IP Cameras, Thermal Cameras, Camera Monitoring, PLC Communication, TCP/IP, Ethernet

Databases: MySQL, PostgreSQL, MongoDB, SQLite

Tools: Git, GitHub, VS Code, Linux Terminal

PROFESSIONAL EXPERIENCE

Computer Vision / AI-ML Engineer Intern
AHOMLAMA

Aug 2026 – Present
On-site

- Working on real-time Computer Vision and Edge AI systems for defence technology applications.
- Contributing to the development and enhancement of an AI-based Aircraft Classification System (ACS) for real-time aircraft detection and classification.
- Working with a multi-stage AI pipeline involving YOLO-based aircraft detection and ConvNeXt-based aircraft classification.
- Contributing to confidence-based validation and out-of-distribution (OOD) detection to improve the reliability of AI predictions.
- Working with day and night datasets for model training, testing, analysis, and performance improvement under varying illumination conditions.
- Deploying and testing AI inference workloads on NVIDIA Jetson edge devices, Linux/Ubuntu systems, and workstations.
- Working with RTSP/IP-stream cameras and thermal cameras for real-time image and video acquisition and computer vision processing.
- Supporting camera monitoring functionality including camera failover, obstruction detection, video recording, logging, and system status monitoring.
- Integrating AI systems with PLC-controlled systems using TCP/IP and Ethernet communication for automated decision and data-transfer workflows.
- Implementing and testing system-to-system communication over Ethernet for transferring logs and operational data between connected workstations.
- Working in a fully offline environment with local datasets, local model inference, local system monitoring, and offline AI deployment without internet dependency.
- Performing debugging, testing, Jetson setup, dataset analysis, documentation, camera validation, PLC integration, and system-level AI integration.

Artificial Intelligence Intern
ADRDE (DRDO), Agra, India

Jan 2026 – Jul 2026
On-site

- Developed and tested computer vision components for defence-related monitoring applications using Python, OpenCV, and image-processing techniques.

- Contributed to a camera-based Aircraft Arrester Barrier Height Measurement System for runway safety applications.
- Implemented ArUco marker-based reference detection and computer vision techniques for estimating barrier net height from long-range camera feeds.
- Developed modular Python components for real-time frame processing, marker detection, measurement calculation, visualization, and warning logic.
- Worked with fixed IP camera configurations and real-time video processing under controlled testing conditions.
- Tested and optimized the system in Linux/Ubuntu environments with focus on reliability, measurement accuracy, and real-time performance.

Summer Research Intern

Jun 2026 – Jul 2026

Indian Institute of Technology (IIT) Jammu

Remote

- Selected for a Summer Research Internship focused on Adversarial Robustness and AI Security.
- Studied GAN-based adversarial attacks and their impact on deep learning image classification systems.
- Researched model extraction and adversarial attack approaches including DFME, Knockoff Nets, JBDA, and related techniques.
- Conducted controlled experiments using local models and open datasets.
- Prepared technical reports, implementation notes, experimental results, and research documentation.

TECHNICAL PROJECTS

Aircraft Classification System (ACS)

Python, YOLO, ConvNeXt, PyTorch, Computer Vision, NVIDIA Jetson, RTSP, Thermal Cameras, PLC, TCP/IP, Ethernet

- Contributing to an AI-based real-time aircraft detection and classification system for defence technology applications.
- Working on a multi-stage vision pipeline using YOLO for aircraft detection followed by ConvNeXt for aircraft classification.
- Contributing to confidence and OOD-based validation for improving prediction reliability and handling uncertain classifications.
- Working with day and night datasets and supporting computer vision processing for varying illumination conditions.
- Supporting real-time deployment and inference on NVIDIA Jetson, Linux/Ubuntu systems, and workstations.
- Working with RTSP/IP-stream cameras and thermal cameras for real-time visual data acquisition and AI processing.
- Supporting camera failover, obstruction detection, video recording, logging, monitoring, PLC communication, and external system integration.
- Working with TCP/IP and Ethernet communication for transferring logs and data between connected systems in a fully offline environment.

Smart Aircraft Arrester Barrier Height Measurement System

Python, OpenCV, ArUco Markers, Computer Vision, RTSP/IP Camera, Linux

- Developed a camera-based computer vision system for monitoring aircraft arrester barrier net height in runway safety applications.
- Implemented ArUco marker-based reference detection for estimating net height from long-range camera video.
- Designed real-time frame-processing and measurement logic for continuous monitoring.
- Implemented warning and visualization logic for detecting potentially unsafe barrier-height conditions.
- Tested the system with IP camera, RTSP, and webcam input in a Linux/Ubuntu deployment environment.

SafeTag QR Emergency Safety Platform

Python, Django, SQLite, Bootstrap, JavaScript, QR Code

[GitHub Repository](#)

- Developed a QR-based emergency safety platform supporting vehicles, children, pets, luggage, and personal safety use cases.
- Built an admin-controlled QR inventory and activation workflow for managing physical QR tags.
- Implemented public QR scanning with emergency contacts, SOS alerts, medical information, and permission-based location sharing.
- Developed dashboard functionality for QR management, activation status, scan logs, and SOS incident tracking.
- Implemented privacy-focused permission workflows for location and evidence collection.

YOLOv5 Web Object Detection Application

Python, YOLOv5, Flask, OpenCV

[GitHub Repository](#)

- Built a Flask-based web application for image-based object detection using YOLOv5.
- Integrated YOLO inference with OpenCV for object detection and bounding-box visualization.

- Developed an interactive interface for uploading images and viewing detection results.

EDUCATION

B.Tech in Computer Engineering

2023 – 2027

Lamrin Tech Skills University, Punjab, India

CGPA: 8+ till 6th Semester

Relevant Coursework: Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Data Structures and Algorithms, Object-Oriented Programming, Python Programming, C++ Programming, Operating Systems, Database Management Systems, Data Analytics, Computer Architecture, Software Engineering

ACHIEVEMENTS

- Selected for Summer Research Internship at Indian Institute of Technology (IIT) Jammu in Adversarial Robustness and AI Security.
- Completed Artificial Intelligence Internship at ADRDE (DRDO), Agra.
- Selected as Computer Vision / AI-ML Engineer Intern at AHOMLAMA for defence technology applications.
- Completed IBM SkillsBuild Cybersecurity Fundamentals Certification.
- Completed Google Cybersecurity Professional Certificate.